IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 10 April 2025, accepted 13 May 2025, date of publication 19 May 2025, date of current version 27 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3571623

RESEARCH ARTICLE

A Smart Web-Based Power Quality and Energy Monitoring System With Enhanced Features

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This work was supported by United Arab Emirates University (UAEU)-Asian Universities Alliance (AUA) Joint Research Fund under Grant 12R130.

ABSTRACT The growing demand for advanced power monitoring underscores the limitations of traditional smart meters, which often lack detailed insights into power quality. This capability has become increasingly important as modern electrical systems are more vulnerable to disturbances and inefficiencies. This paper presents the design and development of a low-cost, real-time power quality and energy monitoring device (PQEm), specifically engineered for easy installation in any electrical setup, whether for bulk metering or submetering of individual units or appliances. The PQEm system integrates a highly precise multiphase analog front-end (AFE) power quality IC with a system-on-a-chip (SoC) platform, enabling the collection of real-time data on key energy metrics such as active, reactive, and apparent power, while also detecting a wide spectrum of power quality disturbances, including sags, swells, interruptions, and voltage imbalances, with a detection latency of less than 5 ms. The PQEm device includes time-frequency analysis to enhance visualization and characterization of voltage and current data. It also supports long-term raw data recording at the native sampling frequency, a feature typically reserved for high-end power quality analyzers. The device has been calibrated and validated through both laboratory testing and real-world applications, offering a cost-effective and user-friendly solution that addresses critical gaps in existing smart metering systems while empowering consumers to monitor power quality issues with greater precision.

INDEX TERMS Power quality monitoring, energy monitoring, power quality, ADE9000, Grafana, Raspberry Pi, InfluxDB.

I. INTRODUCTION

Energy monitoring (EM) and power quality monitoring (PQM) play essential roles in the efficient management of modern electrical systems. While EM focuses on tracking electricity consumption to optimize energy use, PQM evaluates the quality of the electrical supply to ensure that it meets equipment safety and reliability standards. Historically, the primary focus of monitoring systems has been on energy consumption, with less attention paid to power quality (PQ). However, advances in power electronics, shifts in power generation methods, and evolving consumption patterns have highlighted the increasing importance of maintaining high PQ standards.

Common PQ issues, such as voltage fluctuations, harmonic distortions, frequency variations, and voltage sags and swells, can result in serious consequences, including equipment failures, reduced operational lifespans, and costly repairs [1], [2]. These challenges impact not only utilities but also end users, underscoring the need for robust PQ monitoring systems.

In low-voltage electrical systems, PQ issues can significantly degrade the performance, efficiency, and safety of household appliances. Such problems often lead to

The associate editor coordinating the review of this manuscript and approving it for publication was Alexander Micallef.

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malfunctions, premature failures, and increased energy costs [3]. Moreover, many consumers are unaware that household appliances can generate higher levels of total harmonic distortion (THD), further degrading PQ. As modern homes increasingly rely on sensitive electronics and smart devices, monitoring PQ has become more important [4], [5]. Advanced PQM devices are essential for the early detection and resolution of these issues, ensuring infrastructure reliability, extending appliance lifespans, and improving energy efficiency. These devices offer real-time online insights into both the quantity and quality of the power received by end users [6], [7].

The evolution of PQM systems reflects decades of technological progress. Early PQM devices from the 1970s employed basic microprocessors to generate textual reports on PQ events [8], [9]. By the mid-1980s, these systems had evolved to include graphical displays and digital memory, enabling improved data visualization and storage. The 1990s introduced integrated PQM solutions, driven by initiatives such as the Electric Power Research Institute (EPRI) Distribution Power Quality (DPQ) project. These systems included advanced software for data management but faced scalability challenges due to increasing monitoring points and data volume. Modern PQM systems, developed from the 2000s onward, now feature intelligent capabilities, remote monitoring, and sophisticated data analytics.

Previous research on PQM systems has explored various technological approaches, particularly platforms such as microcontrollers, field-programmable gate arrays (FPGAs), and digital signal processors (DSPs). For instance, an FPGA-based system utilizing a Xilinx Virtex-5 and a weighted random vector functional link network classifier was developed for real-time diagnosis of PQ disturbances [10]. Another architecture combined the LabVIEW Real-Time Module with an FPGA processor, employing wavelet analysis for PQ detection [11]. Similarly, a web-enabled PQM system with real-time automation software and FPGA-based monitoring capabilities was proposed for advanced data storage and analysis [12].

DSP-based systems have also been widely adopted. For example, a modular PQM platform built on the NI CompactRIO platform used LabVIEW software for real-time disturbance detection [13]. A notable DSP-based initiative, the I-Grid system, introduced a low-cost, web-enabled PQM solution for industrial facilities across the U.S. and Canada. This system implemented an RMS voltage deviation algorithm to identify voltage anomalies [14]. Additionally, an open three-phase PQ and energy-monitoring device designed for continuous use in industrial facilities, public buildings, and residential homes was proposed in [15], built using a 16-bit analog-to-digital converter (ADC, ADS130E08). More recently, a high-speed DSP-based PQM device was developed using the Hilbert transform and random forest techniques to identify and categorize PQ events [16].

Recently, IoT-based solutions have gained traction in PQ monitoring by providing cloud-integrated, real-time analytics. Reference [17] introduced an Embedded Power Quality Sensor (EPQS) with IoT capabilities, enabling continuous monitoring and automatic detection of PQ disturbances. Reference [18] further expanded this concept with a three-phase smart meter that integrates high-order harmonic detection and cloud-based data processing, offering a cost-effective alternative to traditional PQ analyzers. Cloud-based PQ monitoring systems have also been explored to enhance large-scale data processing. Reference [19] proposed a cloud-integrated PQ analytics platform that uses OpenADR protocols, allowing appliances to dynamically adjust their energy consumption based on real-time grid conditions. Moreover, [20] introduced an IoT-enabled smart meter capable of measuring PQ parameters while integrating a tariff computation model, providing an approach to PQ-based energy pricing.

Another significant advancement in PQ monitoring is the adoption of edge computing, which reduces latency and network congestion by processing data locally. The research presented in [21] developed a low-cost smart meter employing an edge-fog-cloud architecture to detect and classify PQ disturbances efficiently, significantly reducing data transmission requirements while maintaining monitoring accuracy.

Open-source and low-cost PQM solutions have also garnered attention, particularly microcontroller-based systems. For example, an IoT-enabled PQ analyzer was developed using an ATMEGA4809 microcontroller and ESP32 for real-time data monitoring via the ThingSpeak platform, although it lacked features for harmonic analysis [22]. Another study introduced a PQ analyzer using an Arduino Uno and ESP8266 microcontroller, measuring parameters such as RMS voltage, current, power factor, and THD. However, this system did not support the detection of voltage sags and swells [23].

In parallel, advancements in time-frequency analysis techniques, such as the Wavelet Transform (WT) and Stockwell Transform (S-transform), have revolutionized PQ disturbance detection and classification. Unlike traditional time-domain and frequency-domain methods, these techniques provide superior time-localized analysis of transient events. While WT supports multi-scale signal analysis, its performance can be constrained by challenges in selecting the appropriate wavelet and managing resolution [24], [25]. The S-Transform, on the other hand, offers a fully scalable representation with phase information, making it particularly effective in detecting and characterizing various PQ disturbances [26], [27].

The evolution of PQM systems has addressed many challenges in electrical systems, ranging from basic data logging to advanced real-time analytics. However, several limitations remain unaddressed. Existing devices often lack affordability, making them inaccessible for widespread

deployment in low-voltage distribution networks. High-end solutions, such as the Fluke 435 and the Yokogawa CW500, offer comprehensive monitoring capabilities but their high cost restricts their use to industrial settings, limiting broader adoption. Conversely, lower-cost systems frequently compromise on essential features, including precision, time-frequency analysis, high-resolution data storage, and intuitive visualization.

Despite significant advancements in PQM technologies, a notable gap presists in the application of analog front-end (AFE) technologies for low-voltage PQM systems. AFE devices, which integrate high-performance analog-to-digital converters (ADCs) and digital signal processing (DSP) engines, offer substantial advantages over traditional PQM designs [28]. These devices are optimized for specialized applications with well-defined input signal characteristics, dynamic range, and data rate requirements, all while reducing power consumption [29]. The integration of AFE systems into PQM technology represents the next generation of PQM, promising simplified design, reduced costs, and enhanced functionality [30].

This paper addresses critical gaps in power quality monitoring by presenting a novel, low-cost, web-based power quality and energy monitoring (PQEm) system tailored for low-voltage applications. At its core, the system uses the Analog Devices ADE9000, a high-precision multi-phase AFE, integrated with the Raspberry Pi 4 single-board computer SoC. While the ADE9000 serves as the primary sensing element, performing tasks such as signal sampling and fundamental DSP operations including RMS calculation, harmonic analysis, and power factor estimation. However, effective power quality monitoring extends beyond data collection. It requires advanced event characterization, efficient data handling, and user-friendly accessibility. To bridge these gaps, the PQEm system integrates real-time PQ event classification with S-transform analysis, enabling precise time-frequency localization of power disturbances. Additionally, a high-resolution, long-term data storage architecture supports comprehensive historical trend analysis and the identification of recurring issues, which are features often absent in conventional embedded PQ solutions.

The system integrates a custom-designed interface board that facilitates signal conditioning, galvanic isolation, and level shifting, ensuring reliable communication between the AFE and the SoC. It also incorporates a secure, open-source software stack with a Grafana-based dashboard, providing intuitive access to real-time and historical PQ data. This remote visualization capability enhances usability, particularly in distributed energy systems and smart grid applications. Unlike existing solutions that often compromise on cost, performance, or accessibility, the proposed system offers an intelligent and scalable PQ monitoring solution that integrates advanced analytics, real-time processing, and remote accessibility.

The key contributions of this paper are summarized as follows:

1) Design and development of a low-cost, real-time power quality and energy monitoring (PQEm) device that can be easily installed at any electrical installation for bulk metering or for metering/sub-metering of individual units or appliances.
2) Integration of a precise multiphase AFE power quality IC with a SoC platform in the PQEm, enabling real-time monitoring of energy parameters and detection of various PQ disturbances.
3) Implementation of time-frequency analysis using the S-transform to enable detailed visualization and characterization of different disturbances.
4) The ability to record raw data at the native sampling frequency for extended periods, a feature typically found only in sophisticated, high-cost PQ analyzers. This allows for deeper insights and post-processing of PQ events.
5) Development of an open-source software platform for data acquisition, analysis, and visualization, offering users the flexibility to inspect, customize, and extend the system in accordance with evolving PQ standards and individual requirements.

The remainder of this paper is structured as follows: Section II details the hardware and software design of the PQEm system. Section III presents the results and validation of the proposed device through practical testing. Finally, Section IV concludes the study, summarizing the findings and highlighting potential directions for future work.

II. PROPOSED POWER QUALITY AND ENERGY MONITORING DEVICE

In this study, we design, develop, and prototype a Power Quality and Energy Monitoring device, named the ''PQEm''. The PQEm is engineered to monitor PQ events and energy-related parameters in low-voltage distribution networks, in compliance with international standards such as IEC 61000-4-30. The primary function of the device is to provide real-time monitoring and analysis of PQ disturbances, such as voltage sags, swells, harmonics, and flicker, as well as accurate measurement of energy consumption. The PQEm system is an internet-based monitoring and energy metering platform that supports remote operation via Wi-Fi or Ethernet connections, using the integrated networking capabilities of the Raspberry Pi 4 board.

The data collected by the system are accessible through an open-source web application, enabling users to monitor PQ and energy metrics from any web-enabled device. Figure 1 illustrates the overall architecture of the PQEm device, which comprises four main components, the ADE9000 analog front end, a Raspberry Pi single-board computer, InfluxDB for data storage, and a Grafana-based web visualization interface. The ADE9000 serves as the front end, interfacing indirectly

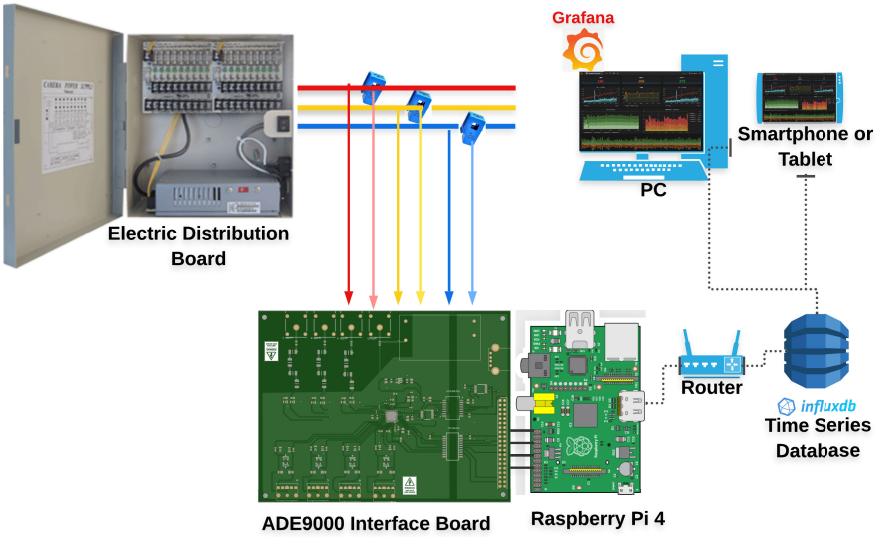


FIGURE 1. Overall architecture of the proposed PQEm device.

with the electrical grid via sensing modules that measure voltage and current waveforms with high precision. The voltage-sensing module employs a voltage divider circuit to reduce the AC voltage to levels suitable for the analog-to-digital converter (ADC). The current-sensing module utilizes an SCT013-000 100 A / 50 mA non-invasive split-core current transformer sensor [31], along with burden resistors. The collected data are processed and analyzed by the Raspberry Pi, which acts as the system's computational core. The device uses specific algorithms to detect and classify PQ events in real-time, making sure that both transient and steady-state anomalies are accurately identified and logged. These algorithms are implemented using a C++ program implemented on the Raspberry Pi, allowing for efficient processing of high-resolution data. Data storage and retrieval are handled by InfluxDB, an open-source time-series database chosen for its ability to manage large volumes of data with high efficiency. The data are stored on a 512 GB solid-state drive (SSD), connected to the Raspberry Pi via a USB 3.0 interface. This configuration not only accelerates data transfer speeds but also improves data reliability and accessibility compared to traditional hard drives, enabling sustained high-rate sampling (up to 32 ksps) without requiring downsampling. Grafana, an interactive web visualization tool, is used to visualize stored data, allowing users to query and display real-time and historical data on PQ and energy consumption. The system architecture is designed so that data visualization is responsive and customizable, providing users with the ability to create detailed dashboards that highlight critical PQ events and energy metrics.

A. HARDWARE PCB DESIGN

The hardware design of the PQEm device was achieved by integrating the ADE9000 IC with a Raspberry Pi 4, which

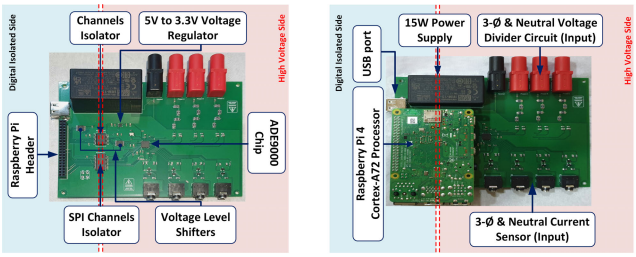


FIGURE 2. Interface PCB board design implementation showing layout of the main components.

features an ARM Cortex-A72 quad-core microprocessor. The ADE9000 is a fully integrated multiphase energy and PQM IC, with high precision. The ADE9000 combines advanced analog capabilities and a DSP core to enable accurate energy monitoring across a wide dynamic range. The internal DSP core performs real-time calculations for critical power parameters, such as RMS values, active and reactive power, and frequency, directly on the chip [32]. This offload of computational tasks reduces the workload on the Raspberry Pi, so that its resources are available to execute algorithms for PQ event detection and classification. For example, the real-time computation of frequency and power measurements by the ADE9000 minimizes overhead, allowing the Raspberry Pi to efficiently handle time-series analysis and classification processes.

Data from the DSP engine within the ADE9000's is extracted via a high-speed Serial Peripheral Interface (SPI) protocol. The chip integrates seven high-performance ADCs, a flexible DSP core, a voltage reference circuit, a temperature sensor, a waveform buffer, and an embedded dsp engine to calculate various electrical parameters. The IC also features an SPI interface and four pulse outputs for interaction with

an external controller. Compared to other energy and PQ monitoring ICs, the ADE9000 offers a high sampling rate of up to 32 kHz and a resolution of 24 bits. By performing the required mathematical calculations internally, the ADE9000 simplifies the design of smart meters and PQM instruments, saving time and reducing costs. The proposed PQEm solution has a total cost of $270, which covers the interface board, Raspberry Pi, SSD storage, as well as auxiliary components like the CTs. This cost is significantly lower compared to other power quality and energy monitoring solutions.

To ensure safe and reliable operation, an interface board was designed for the ADE9000, including all necessary electrical components, as illustrated in Figure 2. This board includes voltage and current input channels, a power supply, and analog-digital isolation circuitry. The current input channels interface directly with current transformers (CTs), which connect via audio jacks on the board to support three-phase and neutral current measurement. A burden resistor across the output of the CTs is used to convert the current output into a measurable voltage. Additionally, fast signal diodes arranged in an anti-parallel configuration provide overvoltage protection by clamping any voltage spikes or transients at the CT output to safe levels.

For the voltage input channels, a voltage divider circuit is used to scale down the input voltage signal so that the voltage amplitude stays within the ADE9000 ADC input limits. The ADE9000's ADCs support fully differential inputs, which enhance noise rejection and accuracy by measuring the difference between two input signals rather than referencing a single-ended signal relative to ground. Accordingly, all voltage and current channels are configured in differential mode. Additionally, digital and analog isolation is essential for reducing noise, preventing ground loops, and protecting the digital controller. The interface board design includes two digital isolators, ADUM4151 and ADUM6404. The ADUM4151 supports high-speed SPI communication up to 17 MHz, providing reliable data transmission between the ADE9000 (analog side) and the Raspberry Pi (digital side). The ADUM6404 is a quad-channel, 5 kV isolator with an integrated DC-to-DC converter, eliminating the need for a separate converter. This isolator is used to isolate the remaining channels that connect the ADE9000 chip to the Raspberry Pi.

To ensure the integrity of signals reaching the ADE9000, antialiasing filters have been included in the interface design for both the current and voltage channels. These filters play a critical role in attenuating high-frequency components that could interfere with the accuracy of analog-to-digital conversion (ADC). The antialiasing filter corner frequency is set around 7 kHz, ensuring sufficient attenuation of unwanted frequencies near the modulator clock frequency of the ADE9000. By filtering the signals before they are digitized, the antialiasing filters enhance the overall precision and reliability of the PQEm device.

Furthermore, voltage level shifting is managed by two level shifter chips, converting voltage levels between 3.3V and 5V as required by the ADE9000 and the isolators, protecting the Raspberry Pi's GPIO pins from potential damage. Moreover, a 15W AC/DC power supply is used to convert the 230V input to a 5V output, capable of delivering up to 3A. This 5V output is accessible via a USB-A port on the PCB. The design provides two options for powering the Raspberry Pi, either directly from the USB-A port on the board or through a power bank with pass-through technology connected to the USB-A port. The pass-through technology allows the power bank to receive 15W of power, enabling it to charge while simultaneously powering another device. This setup works as an uninterrupted power supply (UPS) to the PQEm system, maintaining remote access to the Raspberry Pi and enabling continuous data collection even during extended power outages at the monitoring location. Finally, all ADE9000 channels can be accessed through the Raspberry Pi header, located on the digitally isolated side of the interface board, allowing seamless communication and interaction with the ADE9000.

B. SOFTWARE DESIGN

Software applications play a crucial role in communicating, retrieving, storing, analyzing, and visualizing energy and PQ data.

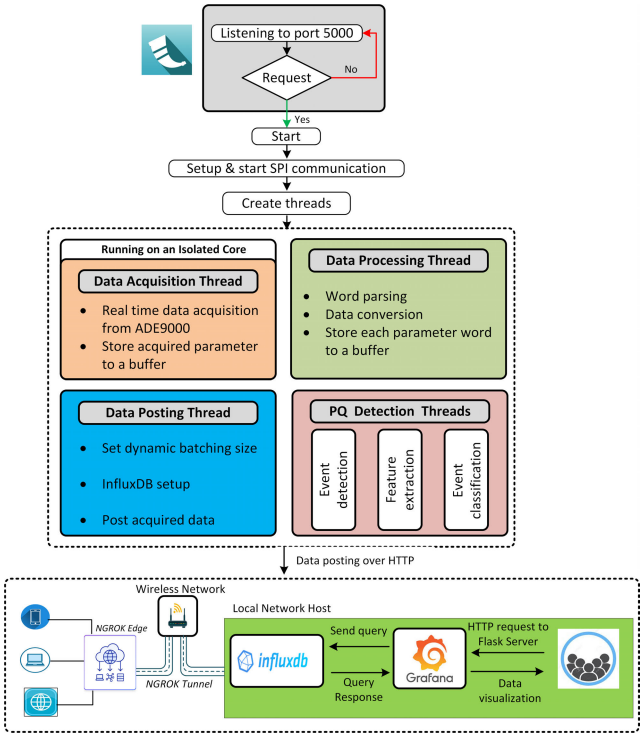


FIGURE 3. Flow chart of PQEm software application.

Figure 3 illustrates the main process flow of the proposed software application for the PQEm device, which is implemented on a Raspberry Pi. The application uses the Flask Python web framework to manage the initiation and termination of the PQEm program, which is visualized using

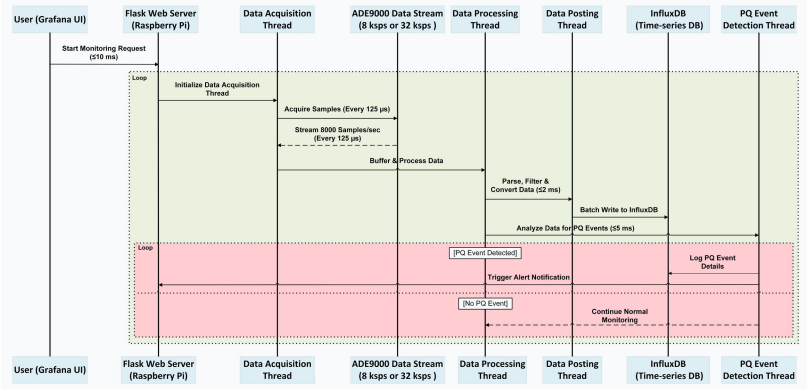


FIGURE 4. Sequence diagram illustrating time durations of key processes in the real-time power quality monitoring system.

Grafana, an open-source interactive web visualization tool. When a user starts the monitoring process from Grafana, the Flask application sends an HTTP request to the Raspberry Pi via Port 5000 to begin PQ event detection and energy monitoring. The program consists of four main parallel threads, as highlighted in Figure 3.

The Raspberry Pi 4B operates on a Linux-based system, which manages CPU time through various scheduling techniques but lacks deterministic timing for real-time tasks. This limitation can result in delays in the acquisition of critical data from the ADE9000 IC, potentially leading to data loss. To address this issue, the Broadcom BCM2711 processor, with its quad-core ARM Cortex-A72 CPU, allows the isolation of one core from the Linux scheduler. By assigning the data acquisition thread to this isolated core using the CPU-affinity Linux API, the system ensures real-time acquisition of waveforms, voltage and current channels, and DSP-calculated values from the ADE9000 chip.

Furthermore, buffering mechanisms are implemented to handle continuous data streams, maintain consistent sampling rates throughout the acquisition and processing stages, and prevent data loss. This ensures uninterrupted data acquisition, which is critical for real-time applications. As demonstrated in Figure 4, the system employs a structured, multi-threaded approach to optimize performance and minimize latency. The Data Acquisition Thread continuously streams samples from the ADE9000 data stream at a rate of 8 kSPS or 32 kSPS (every 125 μs), ensuring high-resolution signal monitoring.

To further enhance efficiency, a separate Data Processing Thread parses, filters, and converts the raw data within $\leq$2 ms, leveraging optimized data handling techniques. The system also dynamically batches processed data before writing it to InfluxDB, significantly reducing database operation overhead. This batch processing approach ensures minimal latency and high-speed data throughput by grouping data points before storage instead of writing each one individually.

Additionally, the PQ Event Detection Thread continuously analyzes the data for power quality (PQ) events within $\leq$5 ms. If an anomaly is detected, an alert notification is triggered, and the Data Posting Thread logs the event details in InfluxDB, as shown in the event detection path in Figure 4. Otherwise, the system continues normal monitoring.

The final stage of the application involves data visualization through Grafana, where users can monitor real-time and historical power quality metrics. For remote access outside the local network, Ngrok provides a secure tunnel, allowing users to access Grafana from any location as if it were hosted on a public server.

1) PQ EVENT DETECTION AND CLASSIFICATION ALGORITHM

In the proposed PQEm system, one of the main processes running on the Raspberry Pi is an algorithm responsible for monitoring, detecting, and classifying PQ disturbances. The PQ events analyzed include voltage sag, voltage swell, momentary interruptions, voltage unbalance, and harmonic distortions. These disturbances are defined according to the IEC 61000-4-30 standard as follows:

- **Voltage Sag**: A reduction in the RMS voltage or current at the power frequency, typically ranging between 0.1 and 0.9 per unit (pu), lasting from 0.5 cycles to 1 minute.
- **Voltage Swell**: An increase in the RMS voltage at the power frequency, typically ranging between 1.1 and 1.8 pu, also lasting from 0.5 cycles to 1 minute.
- **Momentary Interruption**: Complete loss of voltage (below 0.1 pu) on one or more phase conductors, occurring for a duration between 0.5 cycles and 3 seconds.
- **Harmonics**: Frequency components of a periodic signal that are integer multiples of the fundamental frequency (e.g., 50 or 60 Hz) at which the supply system operates.
- **Voltage Unbalance**: A condition in which the deviation of one or more phase voltages from the average of the three-phase system is significant. It is calculated as the ratio of the negative or zero-sequence component to the positive-sequence component, expressed as a percentage. The voltage unbalance percentage [33] can

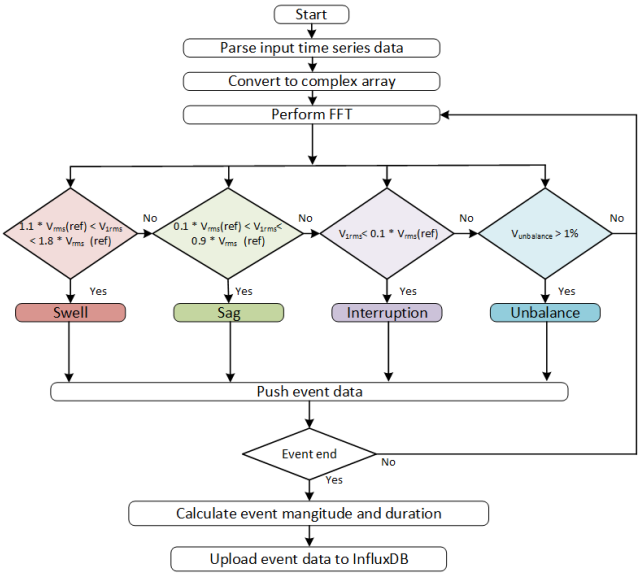


FIGURE 5. PQ event detection and classification algorithm.

be calculated using the following equation:

$$\text{Voltage Unbalance (\%)} = \sqrt{\left(\frac{6 \cdot U_{12}^2 \cdot U_{23}^2 \cdot U_{31}^2}{(U_{12}+U_{23}+U_{31})^2}\right) - 2} \tag{1}$$

The PQ event detection and classification algorithm is designed to identify and classify disturbances in real time. The algorithm demonstrated in Figure 5, continuously monitors the fundamental RMS values of the phase voltages, which are calculated for each half-cycle. The ADE9000 extracts the fundamental frequency component ($V_{1\text{RMS}}$) from a distorted voltage signal by leveraging its internal low-pass filter (LPF). The process begins with the squared voltage signal passing through the LPF, which effectively attenuates harmonics by at least 64 dB, isolating the fundamental frequency component (50 Hz or 60 Hz). After filtering, a square-root algorithm is applied to compute the RMS value accurately. This method ensures that $V_{1\text{RMS}}$ remains unaffected by harmonic distortions, making it highly reliable for power quality monitoring. With a full-scale variation of only $\pm 0.064\%$, the ADE9000 provides precise fundamental RMS measurements, even in environments with significant waveform distortion.

According to the definition of each type of PQ event, specific thresholds are used for the detection and segmentation of these disturbance.

Upon detection of a PQ event, the algorithm initiates a segmentation process. This process involves capturing and storing the voltage data for four cycles preceding the event, the event duration itself, and four cycles following the event. This data provides valuable insight into the nature and potential causes of the disturbance, allowing for more in-depth analysis. The captured data is then uploaded to an InfluxDB database, where it is available for further analysis and visualization. While the primary focus of the algorithm is on transient events such as sags, swells, and interruptions, it also continuously monitors harmonic distortions as a long-term PQ indicator. The harmonic content is analyzed to make sure it meets relevant standards and to detect any deviations that could indicate issues within the electrical system.

2) IMPLEMENTATION OF STOCKWELL TRANSFORM USING COOLEY–TUKEY FFT

The S-transform is a time–frequency analysis technique that combines the strengths of the Short-Time Fourier Transform (STFT) and the wavelet transform (WT). Its ability to provide a frequency-dependent resolution makes it well suited for analyzing non-stationary signals. The S-transform is defined as:

$$S(t,f) = \int_{-\infty}^{\infty} x(\tau)\, \frac{|f|}{\sqrt{2\pi}}\, e^{-\frac{(t-\tau)^2 f^2}{2}}\, e^{-j2\pi f\tau}\, d\tau, \tag{2}$$

where $x(\tau)$ is the time-domain signal, f is the frequency, and t is the time. The term

$$\frac{|f|}{\sqrt{2\pi}}\, e^{-\frac{(t-\tau)^2 f^2}{2}}$$

represents a frequency-dependent Gaussian window. To improve efficiency, the S-transform can be expressed in the frequency domain using the Fourier transform. The process is based on the convolution theorem, which states that multiplication in the frequency domain corresponds to convolution in the time domain. The computational steps can be outlined as follows:

1) Compute the Fourier transform $X(f)$ of the input signal $x(t)$:
$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt \tag{3}$$
2) For each frequency f_k, apply a Gaussian window in the frequency domain:
$$G(f, f_k) = e^{-2\pi^2 \left(\frac{f-f_k}{f_k}\right)^2}, \tag{4}$$
where $G(f, f_k)$ is the Gaussian filter centered at f_k with a standard deviation proportional to f_k.
3) Multiply the Gaussian window $G(f, f_k)$ with the Fourier transform $X(f)$ to localize the frequency content:
$$X_{\text{filtered}}(f, f_k) = X(f) \cdot G(f, f_k) \tag{5}$$
4) Perform the inverse Fourier transform (IFFT) to obtain the localized signal in the time domain for frequency f_k:
$$S(t, f_k) = \int_{-\infty}^{\infty} X_{\text{filtered}}(f, f_k)\, e^{j2\pi ft}\, df \tag{6}$$

The result is a time–frequency representation $S(t,f)$ that provides both time and frequency resolution for the signal $x(t)$.

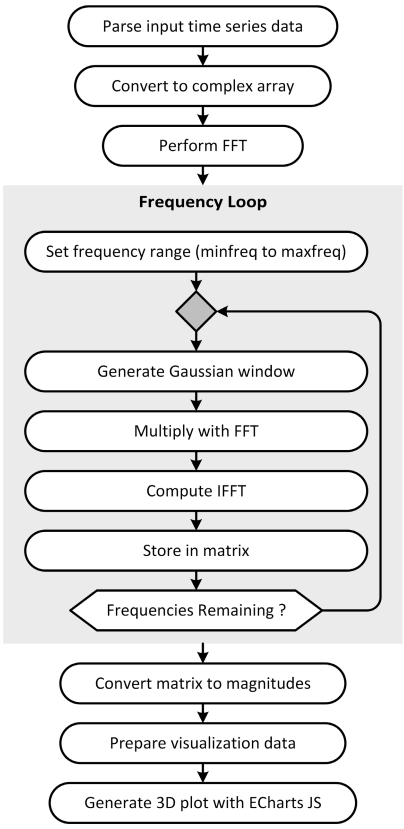


FIGURE 6. Flowchart for implementing the S-Transform using FFT.

The flowchart in Figure 6 outlines the implementation process of the S-transform using the Fast Fourier Transform (FFT). The process begins by parsing the input time series data and converting it into a complex array suitable for FFT computation. Once the FFT is performed, the algorithm enters the Frequency Loop, where a Gaussian window is applied to the signal for each frequency range. The windowed FFT is then transformed back to the time domain using the Inverse FFT (IFFT), and the result is stored in a time-frequency matrix. This loop continues until all specified frequencies are processed. Finally, the matrix is converted to magnitudes, visualization data is prepared, and a 3D time-frequency plot is generated using ECharts JS. This visualization helps to understand the time-frequency characteristics of the input signal.

3) S-TRANSFORM ON EMBEDDED SYSTEMS: COMPUTATIONAL CHALLENGES AND OPTIMIZATIONS

While the S-Transform (ST) provides superior time-frequency resolution, its implementation on embedded hardware presents significant challenges due to its computational intensity. The ST involves applying frequency-dependent Gaussian windows across the entire time series, resulting in a computational complexity of $\mathcal{O}(N^2)$, which is substantially higher than that of the Fast Fourier Transform (FFT), which operates at $\mathcal{O}(N \log N)$. This increased complexity makes the ST particularly demanding in systems with limited processing capabilities.

The transform also requires extensive floating-point multiplications and complex arithmetic, especially when evaluated at each time step and across multiple frequency bins. In the system under consideration, signals are sampled at 8 kSPS, and the ST is performed using a window size of $N = 1024$. Assuming 512 frequency bins (from a 1024-point FFT applied to real-valued signals), the number of complex multiplications per window exceeds 500,000, placing a substantial computational load on the processor.

Memory usage is another key challenge. The S-transform typically produces a time-frequency matrix of size $N \times M$, where N is the number of time samples and M is the number of frequency bins. For $N = 1024$ and $M = 512$, the matrix contains $1024 \times 512 = 524{,}288$ values. If each value is stored in 64-bit floating-point format (8 bytes), the total memory requirement is:

$$1024 \times 512 \times 8 = 4{,}194{,}304 \text{ bytes} \approx 4 \text{ MB}$$

Although the Raspberry Pi 4, equipped with a 1.5 GHz quad-core ARM Cortex-A72 CPU and up to 4 GB of RAM, can accommodate this workload, it is essential to optimize the computations to ensure efficient real-time performance.

All computations were implemented in the `C++` programming language to take advantage of low-level memory management and performance tuning. To further enhance execution speed and maintain system responsiveness, multithreading was employed using POSIX threads. These optimizations enabled the ST to run efficiently in parallel with other critical tasks such as data logging, communication, and web-based monitoring.

III. RESULTS AND DISCUSSION

The results section demonstrates the practical application of the PQEm device in real-world environments. This involves validating the effectiveness of the device for PQ monitoring and energy metering by installing it in residential buildings.

A. CALIBRATION PROCESS FOR THE PQEM DEVICE

Before deploying the PQEm device for real-time data logging and PQM, a comprehensive calibration process was performed to minimize measurement uncertainties and ensure device accuracy. Calibration plays a critical role in controlling errors and uncertainties within measurement systems, aligning the performance of the PQEm device with the standards and ensuring reliable operation under diverse conditions.

In this study, the calibration process was performed using a high-precision experimental setup, using a Fluke 434 PQ Analyzer and a CINERGIA three phase AC grid simulator. The Fluke 434 served as the reference standard for PQ measurements, providing high accuracy and resolution, while the grid simulator enabled precise control over AC voltage waveforms. The simulator introduced user-defined harmonics with a setpoint accuracy of $\pm 0.1\%$, making it ideal

for generating calibration waveforms with controlled levels of distortion and dynamic response.

As demonstrated in Figure 7(a), the calibration setup included the grid simulator to generate specific voltage waveforms and an RL load bank to replicate various load conditions. This combination allowed for the precise calibration of the PQEm's voltage and current measurement channels. The calibration process involved the following steps:

1) VOLTAGE CHANNELS CALIBRATION

The ADE9000's voltage channels were calibrated using the calibration constant xVGAIN, applied as:

$$\text{Voltage Channel Gain} = 1 + \frac{\text{xVGAIN}}{2^{27}} \tag{7}$$

The measured RMS voltage values (V_{Measured}) were compared to the reference voltage values ($V_{\text{Reference}}$) provided by the Fluke analyzer. Any difference between the measured and reference values was addressed by iteratively adjusting the calibration constant xVGAIN, calculated using:

$$\text{xVGAIN} = \text{ROUND}\left((\text{Desired Voltage Gain} - 1) \times 2^{27}\right) \tag{8}$$

This process ensured that the PQEm device accurately measured voltages in the range 0 V to 600 V with an error margin of 0.4%.

2) CURRENT CHANNELS CALIBRATION

The calibration of the current sensors followed a similar methodology, using the calibration constant xIGAIN. The current channel gain was determined by:

$$\text{Current Channel Gain} = 1 + \frac{\text{xIGAIN}}{2^{27}} \tag{9}$$

The calibration constant xIGAIN was adjusted based on the difference between the measured RMS current (I_{Measured}) and the reference current ($I_{\text{Reference}}$) values, as provided by the Fluke analyzer. The iterative adjustment of xIGAIN was conducted using:

$$\text{xIGAIN} = \text{ROUND}\left((\text{Desired Current Gain} - 1) \times 2^{27}\right) \tag{10}$$

This calibration ensured that the PQEm device could accurately measure currents in the range 0 A to 100 A with an error margin of 1%.

3) PHASE CALIBRATION

Phase calibration was performed to correct phase errors between the voltage and current channels. The procedure begins with applying a nominal voltage and current with a lagging 0.5 power factor. The energy registers are then configured to accumulate energy over a 1-second period. After accumulation, the active and reactive energy registers are read to calculate the phase error. The phase calibration register value, APHCAL0, is then determined using the phase error and the following formula:

$$\text{APHCAL0} = \left(\frac{\sin(\varphi - \omega) + \sin\omega}{\sin(2\omega - \varphi)}\right) \times 2^{27} \tag{11}$$

where

$$\omega = 2\pi \times \frac{f_{\text{DSP}}}{f_{\text{LINE}}},$$

with f_{LINE} being the line frequency and f_{DSP} the DSP sampling frequency. The calculated APHCAL0 value is written into the calibration register.

B. VALIDATION OF CALIBRATION AND RESULTS

The calibration process demonstrated that the PQEm device accurately measures voltages in the range of 0 V to 600 V with an error margin of 0.4% and currents from 0 A to 100 A with an error margin of 1%. These values were validated by comparing the PQEm's measurements with those of the Fluke 434 under controlled conditions.

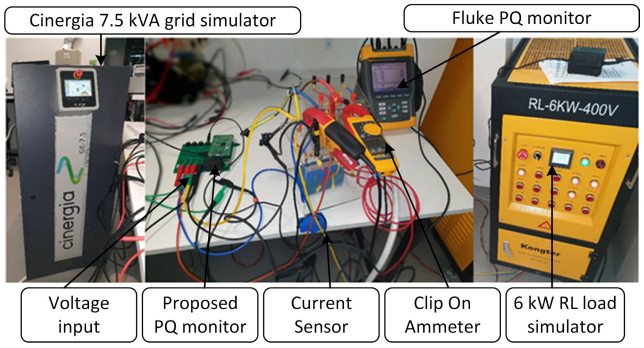
Figure 7(b) illustrates a comparison of the voltage waveforms captured by the PQEm device and the Fluke 434. A marginal error of 0.8% was observed between the two devices, primarily attributed to their sampling rate differences,8 ksps for the PQEm and 7.5 ksps for the Fluke analyzer. To allow direct comparison, the waveform captured by the PQEm device was downsampled to match the Fluke analyzer's sampling rate. Despite this slight mismatch, the PQEm demonstrated a high degree of accuracy in capturing voltage waveforms.

As demonstrated in Figure 7(c) when devices were tested under harmonic distortion conditions with a THD of 50%, both systems provided accurate measurements of the fundamental frequency and third harmonic components. These results validate the PQEm device's capability to effectively capture and analyze voltage waveforms and harmonic distortions, critical for PQ assessment in real-world electrical systems.

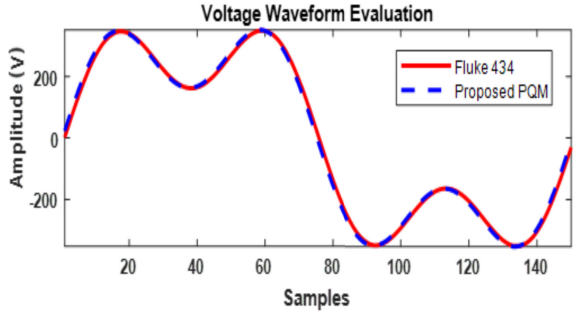
This calibration process established the PQEm device as a reliable tool for power quality and energy monitoring. The rigorous testing with high-precision equipment ensured that the system met high standards of accuracy and reliability, making it suitable for diverse operational scenarios.

C. REAL-WORLD DEPLOYMENT AND DATA ACQUISITION OF THE PQEM SYSTEM

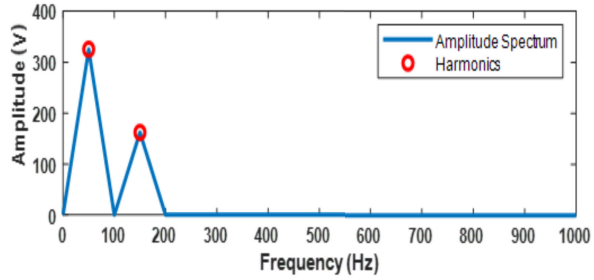
To assess the performance of the designed PQM system in a real-world setting, the device was installed and tested at a residential site in Al Ain City, United Arab Emirates. Measurements were conducted over a three-week period. All collected data were stored and processed at the native sampling frequency of the ADC, with no loss of resolution. Data access was efficient thanks to the speed of the SSD storage and the reliability of InfluxDB. Figure 8(a) illustrates the connection of the PQEm device to the electrical panel.



(a)



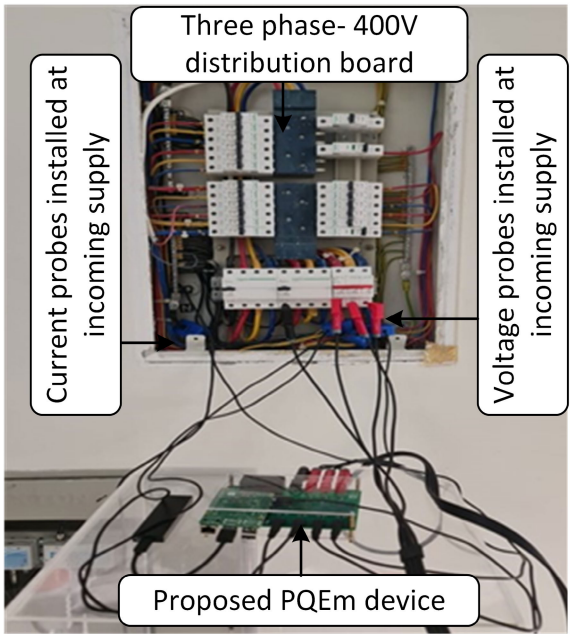
(b)



(c)

FIGURE 7. Calibration and performance evaluation of the PQEm device. (a) Experimental setup using a grid simulator and RL load bank. (b) Comparison of voltage waveforms captured by the PQEm and Fluke 434. (c) Measurement under 50% THD conditions, confirming accurate detection of fundamental and third harmonic components.

The data collected by the PQEm device is visualized using the open-source Grafana platform, with InfluxDB as the data source. The PQEm dashboard features four primary panels: waveform scope, PQ monitoring, energy monitoring, and harmonic analysis, as shown in Figure 8(b). Through this dashboard, users can conveniently access all data from the ADE9000 IC. Monitoring process can be started and stopped remotely using the main button on the dashboard, with the flexibility to monitor one or all three phases simultaneously. The main panel displays real-time voltages, currents, and RMS values, with an adjustable update rate ranging from 0.5 to 3 seconds. Additionally, the sampling rate can be set to either 8 ksps or 32 ksps, the latter being the maximum provided by the ADE9000.



(a)

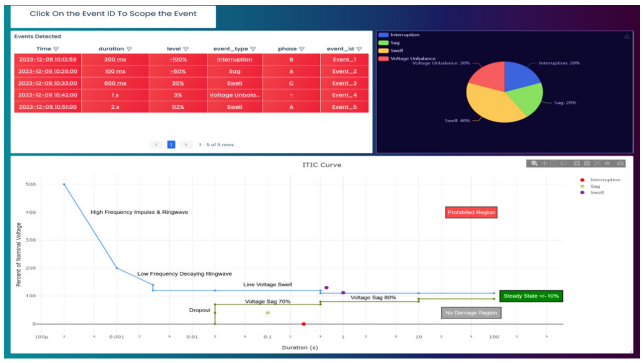


(b)

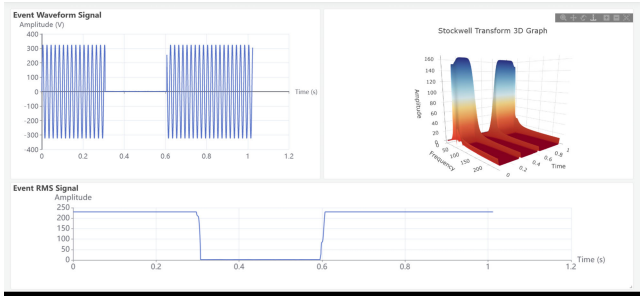
FIGURE 8. Real-world deployment and interface of the PQEm system. (a) PQEm device installed in the electrical panel of a residential building in Al Ain, UAE. (b) PQEm dashboard in Grafana displaying waveform scope, with real-time data visualization and remote control capabilities.

D. POWER QUALITY MONITORING WITH ENHANCED EVENT ANALYSIS AND VISUALIZATION USING TIME-FREQUENCY TECHNIQUES

Figure 9(a) displays the PQEm device dashboard in Grafana, showing the PQ events recorded during various PQ disturbances. This panel includes a table listing detected events with an insights into each disturbance, such as type, duration, magnitude, and the affected phase. Alongside the



(a)



(b)

FIGURE 9. Power quality event detection and analysis using the PQEm system. (a) Grafana PQ Monitoring Panel displaying recorded events with statistical summaries, ITIC curve visualization, and real-time alerting. (b) 3D time-frequency plot of a PQ event using the S-transform, highlighting time and frequency localization.

event table, a pie chart summarizes the event statistics. The Grafana dashboard is also configured to send email alerts upon detection of specific disturbances, attaching a screenshot of the PQ panel. This feature keeps users informed of detected events or anomalies in real time. Additionally, the recorded PQ disturbances can be visualized on the Information Technology Industry Council (ITIC) curve, which outlines the acceptable voltage range that sensitive electronic equipment can tolerate without malfunction or damage. As shown on the ITIC curve, voltage sags and interruptions could potentially cause sensitive equipment to malfunction, while voltage swells might lead to equipment failure.

Furthermore, users can further investigate the event signals by selecting them from the event table, allowing for the examination of pre-event and post-event conditions once voltage returns to normal.

E. TIME-FREQUENCY ANALYSIS OF POWER QUALITY DISTURBANCES USING THE S-TRANSFORM

The S-Transform is a powerful time-frequency analysis tool that enhances the detection and classification of power quality disturbances by providing a multi-resolution spectral representation. Unlike traditional Fourier-based methods, which only analyze signals in the frequency domain, the S-Transform preserves both time and frequency information. This makes it highly effective for tracking both transient and steady-state disturbances in power systems. The accuracy and resolution of time-frequency analysis using the S-Transform are directly influenced by the sampling rate. A higher sampling rate provides better frequency resolution, enabling more precise detection of transient power quality disturbances. However, it increases computational complexity, which is a critical factor for low-cost power monitoring applications. In this study, the power signal was sampled at 8 kSPS, ensuring that both low and high-frequency components of power quality disturbances were captured. To balance time and frequency resolution, an FFT size of N=1024 was selected. This choice provides a frequency resolution of 7.81 Hz, calculated as:

$$\Delta f = \frac{f_s}{N} = \frac{8000}{1024} \approx 7.81 \text{ Hz}$$

The Radix-2 Cooley-Tukey FFT algorithm is utilized for efficient spectral decomposition, allowing for fast computation by breaking down the FFT operation into smaller stages. This step converts the time-domain signal into its frequency-domain representation, enabling further spectral analysis. To improve time localization, a frequency-dependent Gaussian window function is applied to each frequency, as defined in Equation (4). The windowing function smooths the spectral content, reducing spectral leakage and enhancing localization in the time-frequency domain. After windowing, an Inverse FFT is applied to reconstruct the localized time-domain components. This step ensures that frequency components are correctly mapped back to their respective time instances. The output of the S-Transform is a time-frequency matrix, where each element represents the signal's amplitude at a specific time and frequency. This matrix is visualized as a 3D spectrogram, where:

- The X-axis represents time,
- The Y-axis represents frequency,
- The colour intensity represents the magnitude of the signal at a given time and frequency.

This representation enables clear identification of various power quality disturbances, such as voltage sags, harmonics, and transient oscillations. Figure 9(b) shows how the S-Transform captures events using both time and frequency. The chart illustrates that high-frequency signals mainly appear at the beginning and end of the interruptions. This indicates that the S-Transform can clearly show short, sudden changes. The dominant 50 Hz frequency component remains clearly identifiable, demonstrating the effectiveness of the S-Transform in accurately preserving fundamental frequency characteristics essential for reliable power quality assessment.

F. ENERGY MONITORING ASSESSMENT

The developed PQEm device not only provides detailed PQ information, but also offers comprehensive insights into energy consumption, including real, reactive, and apparent power, along with other important parameters. The energy monitoring feature is accessible through a user-friendly

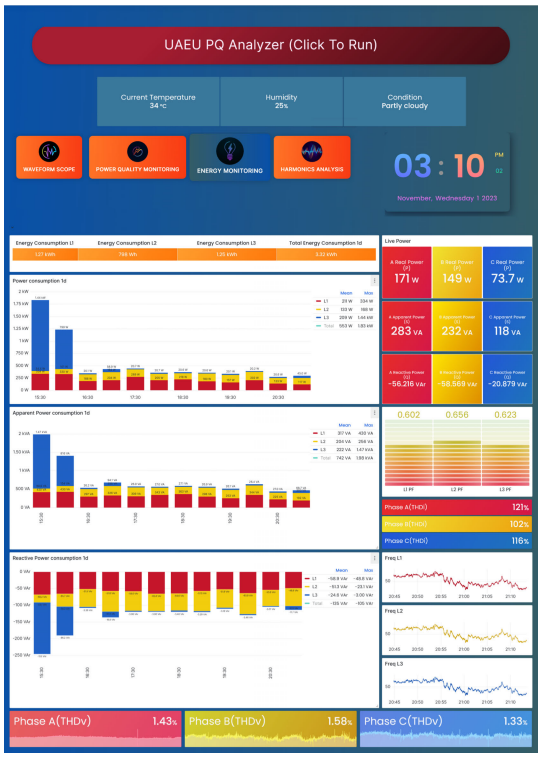


FIGURE 10. Energy monitoring panel indicating various energy related parameters.

interface on the Grafana dashboard, providing a comprehensive overview of both real-time and historical data.

The energy monitoring dashboard, shown in Figure 10, is divided into several key sections that display energy-related data in real time. At the top, weather data for the monitoring location is displayed, sourced dynamically from WeatherAPI. By integrating real-time weather data into the energy monitoring dashboard, the PQEm system allows users to assess how environmental conditions affect energy usage. For example, during the hot summer days in Al Ain City, the system recorded a significant increase in energy consumption due to the prolonged operation of air conditioning units. The correlation between external temperature and energy use highlights the importance of adjusting consumption patterns based on weather conditions. By understanding these relationships, users can adopt energy-saving measures, such as increasing thermostat settings or using energy-efficient cooling devices.

Additionally, the main energy metrics displayed on the dashboard include active power (in kilowatts), reactive power, and apparent power, with real-time updates every 30 seconds. The system also provides key metrics such as power factor and supply frequency, enabling users to track how efficiently their electrical system is operating. These parameters are important for identifying inefficiencies in power use. For example, a low power factor may indicate the need for power factor correction.

Users can select between different time intervals (daily, weekly, or monthly) to observe trends and variations in energy usage. This allows for the identification of high-consumption periods, offering an opportunity to shift energy-intensive tasks to off-peak times, thereby reducing energy costs. Moreover, a significant feature of the energy monitoring system is its ability to store historical energy data for detailed analysis. The energy dashboard provides visualizations of daily load profiles, with breakdowns of energy usage in hourly intervals. This data is presented in bar charts, making it easy to identify consumption peaks, which can be further analyzed to understand the causes of excessive energy use.

For example, during the three-week test period, the PQEm system recorded spikes in energy usage during daytime hours, which coincided with peak air conditioning use in the monitored residential building. The stored data can be exported in spreadsheet format, allowing users to perform more advanced analyses, such as year-on-year comparisons or integration with other energy management systems. This feature makes the system highly adaptable for both residential users and small commercial enterprises aiming to improve their energy efficiency.

In addition to monitoring traditional energy metrics, the PQEm device provides insights into harmonic distortion, which can affect the overall efficiency of electrical systems. The dashboard includes visualizations of both total harmonic distortion in voltage (THDv) and current (THDi). Monitoring these parameters is essential for identifying potential sources of energy loss due to poor PQ.

The PQEm system also features real-time power factor (PF) monitoring, an essential metric for evaluating energy efficiency, where the PF is derived as:

$$\mathrm{PF} = \frac{\text{Watt accumulated over 1.024 s}}{\text{VA accumulated over 1.024 s}} \tag{12}$$

The accumulated active power over a 1.024-second window is divided by the accumulated apparent power over the same period to obtain the power factor (PF). This method ensures high accuracy and stability by averaging out transient fluctuations and noise. PF trends are displayed on the dashboard, enabling users to detect inefficiencies, such as elevated reactive power, and implement necessary adjustments. Consistently low power factor levels may indicate the need for power factor correction devices to minimize losses and enhance energy performance.

G. COMPARISON OF THE PQEM DEVICE WITH OTHER POWER QUALITY ANALYZERS SOLUTIONS

To validate the proposed device, the features and functionalities of the PQEm device were compared with those of various PQ analyzers, such as Fluke 435, Yokogawa CW500 and Open Power Quality. As indicated in Table 1, PQEm outperforms some well-known PQ monitoring devices, being the only one that has high storage capacity, and performs time–frequency analysis and email alert notifications when an event occurs. In addition, the PQEm device is a highly

TABLE 1. Comparison of various devices.

Features	PQEm Device	Fluke 435 [34]	Yokogawa CW500 [35]	Open Power Quality [36]
Multiphase Capability	Yes	Yes	Yes	No
Email Alert	Yes	No	No	No
User Interface	Web App	Display	Display	Web App
Storage Capacity	512 GB SSD	8 GB	2 GB SD	Cloud Storage
Database Structure	InfluxDB - (Time series Database)	-	-	MongoDB – (NoSQL Database)
Time-Frequency Analysis	Yes	No	No	No
Long-Term waveform Recording	Yes	Yes	No	No
Communication & Connectivity	Ethernet/ Wi-Fi	Optical cable	USB communication/ Bluetooth	Wi-Fi
Price	Low	High	High	Low

cost-effective solution for both PQ and EM compared to commercially available PQ analyzers.

IV. CONCLUSION

This research successfully developed a low-cost, real-time (PQEm) system designed for residential applications, bridging the gap between consumer accessibility and advanced PQ monitoring. By integrating a high-precision multiphase (AFE) with a Raspberry Pi single-board computer, the PQEm system offers a versatile solution capable of detecting and analyzing a wide spectrum of PQ disturbances, such as voltage sags, swells, and harmonics, while also monitoring critical energy parameters in real-time. The system's open-source architecture allows users to customize features and adapt to changing PQ standards. Its Grafana-powered interface provides an intuitive, web-based platform for both real-time and historical data visualization, enabling end-users to easily monitor their electrical systems. The combination of affordability and the capability to store and process large volumes of data at high sampling rates positions PQEm as an ideal tool for residential and small-scale commercial applications. It offers significant advantages over existing solutions in terms of accessibility, ease of installation, and the comprehensiveness of the data provided. The PQEm system empowers users to proactively manage their energy consumption and address PQ issues, contributing to more reliable and efficient electrical systems. To further enhance the system's responsiveness and adaptability, future work will explore the integration of adaptive filtering techniques. Methods such as Least Mean Squares (LMS) and Recursive Least Squares (RLS) offer real-time tracking capabilities by continuously adjusting filter parameters in response to evolving signal conditions. These techniques are particularly beneficial for detecting and tracking time-varying disturbances, such as slowly increasing harmonics, flicker, or transients. Additionally, the integration of non-intrusive load monitoring (NILM) techniques could significantly expand the device's capabilities by enabling the disaggregation of total energy consumption into appliance-level data. This would provide users with deeper insights into their energy usage pattern without requiring additional hardware.

REFERENCES

[1] M. H. J. Bollen, "What is power quality?" *Electric Power Syst. Res.*, vol. 66, no. 1, pp. 5–14, Jul. 2003.

[2] S. Chattopadhyay, M. Mitra, and S. Sengupta, *Electric Power Quality* (Power Systems). Dordrecht, The Netherlands: Springer, 2011.

[3] P. Sanjeevikumar, *Power Quality in Modern Power Systems*. London, U.K.: Academic Press, 2021.

[4] A. Grevener, J. Meyer, S. Rönnberg, M. Bollen, and J. Myrzik, "Survey of supraharmonic emission of household appliances," *CIRED-Open Access Proc. J.*, vol. 2017, no. 1, pp. 870–874, Oct. 2017. [Online]. Available: https://digital-library.theiet.org/content/journals/10.1049/oap-cired.2017.0458

[5] G. Ye, M. Nijhuis, V. Cuk, and J. F. G. Cobben, "Stochastic residential harmonic source modeling for grid impact studies," *Energies*, vol. 10, no. 3, p. 372, Mar. 2017. [Online]. Available: https://www.mdpi.com/1996-1073/10/3/372

[6] *Electromagnetic Compatibility (EMC)—Part 4-30: Testing and Measurement Techniques-power Quality Measurement Methods*, Standard IEC 61000-4-30, Int. Electrotechnical Commission, 2015. [Online]. Available: https://webstore.iec.ch/en/publication/68642

[7] X. Liang, "Emerging power quality challenges due to integration of renewable energy sources," *IEEE Trans. Ind. Appl.*, vol. 53, no. 2, pp. 855–866, Mar. 2017.

[8] R. C. Dugan, M. F. McGranaghan, S. Santoso, and H. W. Beaty, *Electrical Power Systems Quality*. New York, NY, USA: McGraw-Hill, 2012.

[9] A. Dranetz. (2024). *Abe Dranetz–Dranetz Power Monitoring*. [Online]. Available: https://www.dranetz.com/abe-dranetz/

[10] M. Sahani and P. K. Dash, "FPGA-based online power quality disturbances monitoring using reduced-sample HHT and class-specific weighted RVFLN," *IEEE Trans. Ind. Informat.*, vol. 15, no. 8, pp. 4614–4623, Aug. 2019.

[11] A. Delle Femine, D. Gallo, C. Landi, and M. Luiso, "Power-quality monitoring instrument with FPGA transducer compensation," *IEEE Trans. Instrum. Meas.*, vol. 58, no. 9, pp. 3149–3158, Sep. 2009.

[12] O. Yildirim, B. Eristi, H. Eristi, S. Unal, Y. Erol, and Y. Demir, "FPGA-based online power quality monitoring system for electrical distribution network," *Measurement*, vol. 121, pp. 109–121, Jun. 2018.

[13] M. V. E. Xavier, W. D. C. Boaventura, and M. D. Noronha, "High performance power quality monitoring system," in *Proc. 17th Int. Conf. Harmon. Quality Power (ICHQP)*, Oct. 2016, pp. 974–979.

[14] D. Divan, G. A. Luckjiff, W. E. Brumsickle, J. Freeborg, and A. Bhadkamkar, "A grid information resource for nationwide real-time power monitoring," *IEEE Trans. Ind. Appl.*, vol. 40, no. 2, pp. 699–705, Mar. 2004.

[15] L. Morales-Velazquez, R. D. J. Romero-Troncoso, G. Herrera-Ruiz, D. Morinigo-Sotelo, and R. A. Osornio-Rios, "Smart sensor network for power quality monitoring in electrical installations," *Measurement*, vol. 103, pp. 133–142, Jun. 2017.

[16] M. Sahani, S. Choudhury, S. Kumar Rout, and D. Amaresh Gadanayak, "DSP based online power quality events detection and classification using Hilbert Huang transform and random forest method," in *Proc. Int. Conf. Comput. Intell. Smart Power Syst. Sustain. Energy (CISPSSE)*, Jul. 2020, pp. 1–6.

[17] R. Medina-Gracia, A. Del Rocio Del Rocio Gil de Castro, J. Garrido-Zafra, A. Moreno-Munoz, and E. Canete-Carmona, "Power quality sensor for smart appliance's self-diagnosing functionality," *IEEE Sensors J.*, vol. 19, no. 20, pp. 9486–9495, Oct. 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8746669/

[18] J. Garrido-Zafra, A. R. Gil-de-Castro, R. Savariego-Fernandez, M. Linan-Reyes, F. Garcia-Torres, and A. Moreno-Munoz, "IoT cloud-based power quality extended functionality for grid-interactive appliance controllers," *IEEE Trans. Ind. Appl.*, vol. 58, no. 3, pp. 3909–3921, May 2022. [Online]. Available: https://ieeexplore.ieee.org/document/9737374/

[19] J. Budhavarapu, H. Kumar Singh, K. Thirumala, and M. Sahoo, "Design and implementation of smart meter for PQ-based tariff computation for LV distribution network consumers," *Measurement*, vol. 216, Jul. 2023, Art. no. 112959. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0263224123005237

[20] E. Viciana, F. M. Arrabal-Campos, A. Alcayde, R. Baños, and F. G. Montoya, "All-in-one three-phase smart meter and power quality analyzer with extended IoT capabilities," *Measurement*, vol. 206, Jan. 2023, Art. no. 112309. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0263224122015056

[21] W. L. Rodrigues Junior, F. A. S. Borges, A. F. D. S. Veloso, R. D. A. L. Rabelo, and J. J. P. C. Rodrigues, "Low voltage smart meter for monitoring of power quality disturbances applied in smart grid," *Measurement*, vol. 147, Dec. 2019, Art. no. 106890. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S026322411930747X

[22] G. M. Madhu, C. Vyjayanthi, and C. N. Modi, "Design and development of a novel IoT based smart meter for power quality monitoring in smart grid infrastructure," in *Proc. IEEE Region 10 Conf. (TENCON)*, Oct. 2019, pp. 2204–2209.

[23] L. A. Kumar, V. Indragandhi, R. Selvamathi, V. Vijayakumar, L. Ravi, and V. Subramaniyaswamy, "Design, power quality analysis, and implementation of smart energy meter using Internet of Things," *Comput. Electr. Eng.*, vol. 93, Jul. 2021, Art. no. 107203.

[24] M. A. S. Masoum, S. Jamali, and N. Ghaffarzadeh, "Detection and classification of power quality disturbances using discrete wavelet transform and wavelet networks," *IET Sci., Meas. Technol.*, vol. 4, no. 4, pp. 193–205, Jul. 2010. [Online]. Available: https://digital-library.theiet.org/content/journals/10.1049/iet-smt.2009.0006

[25] M. Uyar, S. Yildirim, and M. T. Gencoglu, "An effective wavelet-based feature extraction method for classification of power quality disturbance signals," *Electr. Power Syst. Res.*, vol. 78, no. 10, pp. 1747–1755, Oct. 2008. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0378779608000953

[26] T. Zhong, S. Zhang, G. Cai, Y. Li, B. Yang, and Y. Chen, "Power quality disturbance recognition based on multiresolution S-transform and decision tree," *IEEE Access*, vol. 7, pp. 88380–88392, 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8746138/?arnumber=8746138

[27] F. Guo, P. Wang, Y. Wang, P. Ren, and Y. Zhang, "Research on improved s transform for the feature extraction of power quality disturbance signal," *IEEE Access*, vol. 8, pp. 137910–137917, 2020. [Online]. Available: https://ieeexplore.ieee.org/document/9149889/?arnumber=9149889

[28] V. V. de Araujo, E. F. Simas Filho, A. Oliveira, W. L. A. de Oliveira, and L. T. C. Esteves, "Integrated circuit for real-time poly-phase power quality monitoring," *Microelectron. J.*, vol. 79, pp. 57–63, Sep. 2018.

[29] S. Bhunia, S. J. Majerus, and M. Sawan, *Implantable Biomedical Microsystems: Design Principles and Applications* (Micro & nano technologies series). Boston, MA, USA: Elsevier, 2015.

[30] N. Chandrappa and S. Banerjee. (2024). *The Next Generation of Power Quality Monitoring Technology—Helping Industrial Equipment Stay Healthy | Analog Devices*. [Online]. Available: https://www.analog.com/en/signals/thought-leadership/the-next-generation-of-power-quality-monitoring-technology.html

[31] YHDC. (2024). *SCT013_dechang Electronic Co., Ltd*. [Online]. Available: http://en.yhdc.com/product1311.html?productId=703

[32] D. Analog. (2024). *ADE9000 High Performance, Multiphase Energy, and Power Quality Monitoring IC*. Analog Devices. [Online]. Available: https://www.analog.com/media/en/technical-documentation/data-sheets/ADE9000.pdf

[33] (2019). *IEEE Recommended Practice for Monitoring Electric Power Quality*. [Online]. Available: https://ieeexplore.ieee.org/document/8796486/

[34] *Fluke 434 / 435 Power Quality and Energy Analyzers*. Accessed: Apr. 4, 2024. [Online]. Available: https://www.fluke.com/en-us/product/electrical-testing/power-quality/434-435?srsltid=AfmBOooweR4hEWqm9qOwO7WgJ0jaES23gNbjI6DYqP3wluqxbecYyPL

[35] Meas. Corp. *CW500–Power Quality Analyzer | Yokogawa Test& Measurement Corporation*. Accessed: Apr. 4, 2024. [Online]. Available: https://tmi.yokogawa.com/eu/solutions/products/portable-and-bench-instruments/portable-power-monitorrecorder/cw500-power-quality-analyzer/

[36] Open Power Qual. *Open Source Hardware and Software for Low-cost Distributed Power Quality Data Collection, Analysis, and Visualization*. Accessed: Apr. 4, 2024. [Online]. Available: https://openpowerquality.org/



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